

HAT-P-19b

R-1303

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_231009 · created 2026-07-31T17:31:57+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460226.79487 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1828 +/- 0.0083
Transit depth (Rp/Rs)^2	3.34 +/- 0.3 [%]
Orbital Inclination (inc)	86.63 +/- 0.17
Airmass coefficient 1 (a1)	1.213 +/- 0.02
Airmass coefficient 2 (a2)	-0.179 +/- 0.05
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#2 - [206, 446]
Optimal Aperture	4.21
Optimal Annulus	13.1
Transit Duration (day)	0.0921 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1828 ± 0.0083 vs 0.1418 (deviation 3.19σ)
Duration fit vs published	0.0921 d vs 0.1196 d (deviation 4.33σ)
Mid-time O–C	12.1 min
Depth SNR	14.2
Residual scatter	1.6 %

Light curve

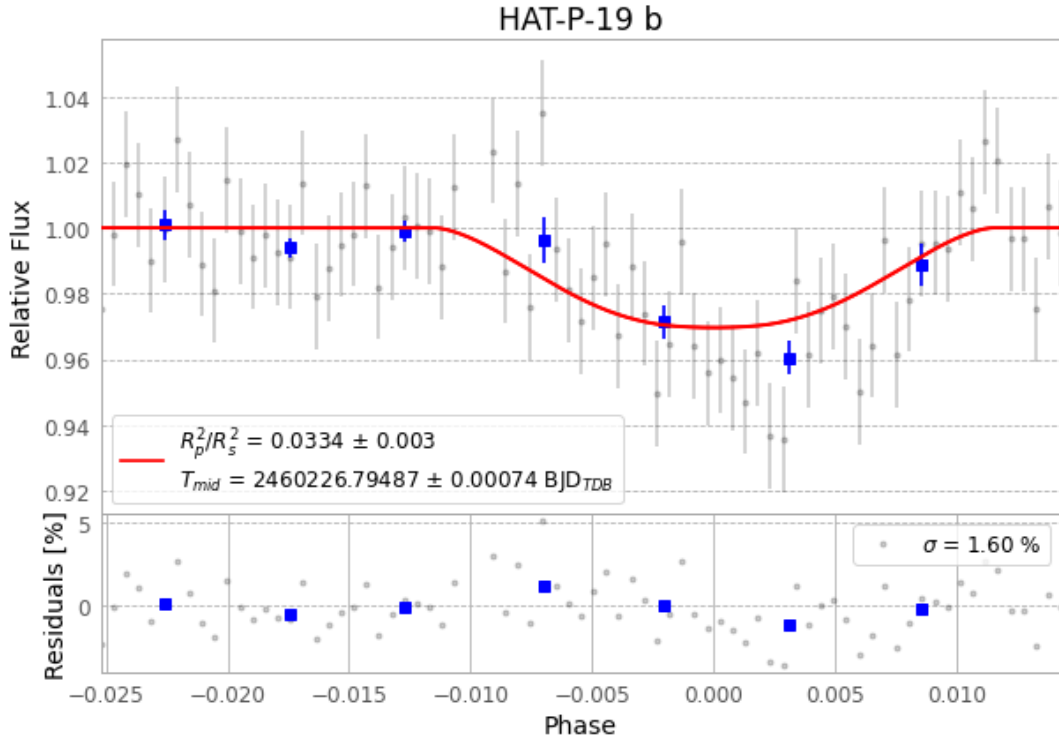


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-10-08.png (SHA-256 bb155f87b6c9481d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9242438e00e2b169...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), HATP-19231009043015.FITS → HATP-19231009081815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-231009.log (4623e56feb7...), FinalLightCurve_HAT-P-19 b_2023-10-08.png (bb155f87b6c9...), FinalLightCurve_HAT-P-19 b_2023-10-08.csv (9d951ea16930...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)